1. OSI Model

The OSI (Open Systems Interconnection) model has 7 layers:

Physical Layer – Transmission of raw bits (Example: Ethernet cables)
Data Link Layer – Frame transmission (Example: MAC address, Switch)
Network Layer – Routing packets (Example: IP, Router)
Transport Layer – End-to-end communication (Example: TCP, UDP)
Session Layer – Session management (Example: NetBIOS)
Presentation Layer – Data formatting (Example: SSL/TLS)
Application Layer – User interaction (Example: HTTP, FTP)

TCP operates at the Transport Layer

2. IP Addressing
Public IP: Accessible over the internet
Private IP: Used within local networks

Private IP ranges:

Class A: 10.0.0.0 – 10.255.255.255
Class B: 172.16.0.0 – 172.31.255.255
Class C: 192.168.0.0 – 192.168.255.255

192.168.1.100 is a Private IP

3. Protocols

Port mapping:

22 → SSH
80 → HTTP
443 → HTTPS
3306 → MySQL

TCP vs UDP:

TCP → Reliable, connection-oriented
UDP → Faster, connectionless

NAT (Network Address Translation):

Converts private IP to public IP
Used to save IP addresses and improve security

```
kali@kali: ~/Desktop/assignment4
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~/Desktop]
└─$ cd assignment4

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ code .

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ nano section3_networking.docx

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:6a:7a:62 brd ff:ff:ff:ff:ff:ff
    inet 192.168.234.128/24 brd 192.168.234.255 scope global dynamic noprefixroute eth0
       valid_lft 1237sec preferred_lft 1237sec
    inet6 fe80::fa04:d434:c53b:b7c2/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

1. Network Configuration (ip a)
IP Address: 192.168.234.128
Interface: eth0
MAC Address: 00:0c:29:6a:7a:62

```
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2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 00:0c:29:6a:7a:62 brd ff:ff:ff:ff:ff:ff
    inet 192.168.234.128/24 brd 192.168.234.255 scope global dynamic nopref
       valid_lft 1237sec preferred_lft 1237sec
    inet6 fe80::fa04:d434:c53b:b7c2/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ ip route
default via 192.168.234.2 dev eth0 proto dhcp src 192.168.234.128 metric 10
192.168.234.0/24 dev eth0 proto kernel scope link src 192.168.234.128 metri

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ cat /etc/resolv.conf
# Generated by NetworkManager
```
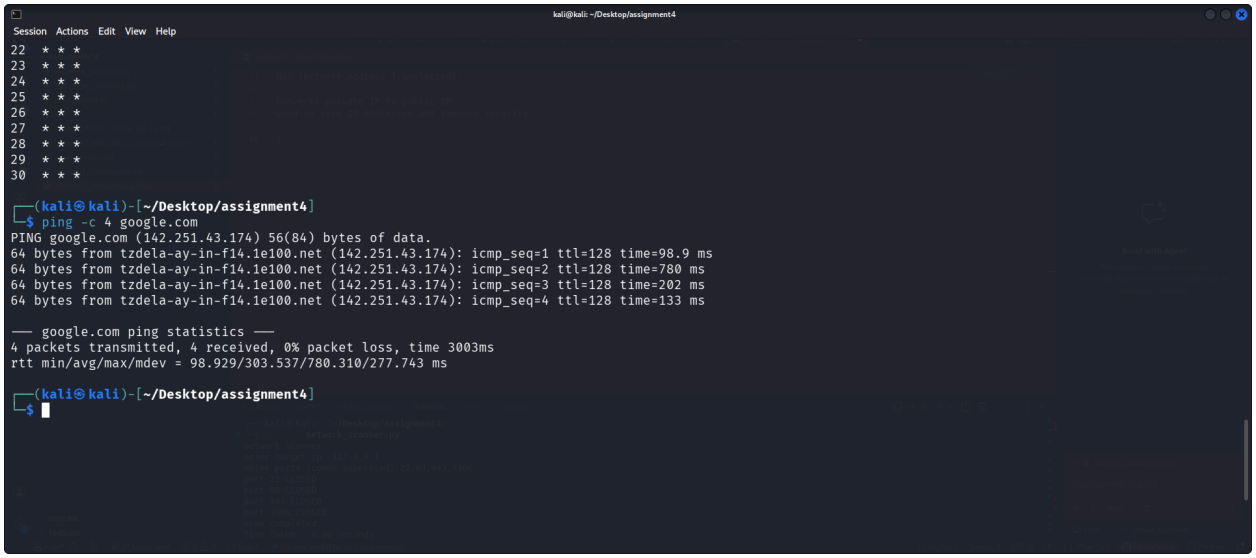

2. Routing Table (ip route)
Default Gateway: 192.168.234.2
Network Range: 192.168.234.0/24

```
kali@kali: ~/Desktop/assignment4
Session  Actions  Edit  View  Help
default via 192.168.234.2 dev eth0 proto dhcp src 192.168.234.128 metric 10
192.168.234.0/24 dev eth0 proto kernel scope link src 192.168.234.128 metri

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.234.2

┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ traceroute google.com
traceroute to google.com (142.251.43.174), 30 hops max, 60 byte packets
 1  192.168.234.2 (192.168.234.2)  3.629 ms  3.270 ms  3.080 ms
 2  * * *
 3  * * *
```

3. DNS Configuration
DNS Server: 192.168.234.2

```
kali@kali: ~/Desktop/assignment4
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~/Desktop/assignment4]
└─$ traceroute google.com
traceroute to google.com (142.251.43.174), 30 hops max, 60 byte packets
 1  192.168.234.2 (192.168.234.2)  3.629 ms  3.270 ms  3.080 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
```

4. Ping Test
Successfully pinged google.com
0% packet loss
Average response time ≈ 303 ms



5. Traceroute Analysis
First hop: 192.168.234.2 (gateway)
Further hops are hidden (* * *) due to network restrictions
Number of hops (visible): 1